Candidate's Assessment Number ________________

CHRISTIAN SOCIAL SERVICES COMMISSION (CSSC)

CSSC JOINT EXAMINATION FOR CHURCH SCHOOLS

EASTERN ZONE

VOCATIONAL STREAM

FORM II PRE – NATIONAL ASSESSMENT

MATHEMATICS

CODE: 043

Time: 2:30 Hours **Year : 2025**

Instructions

1. This paper consists of **ten (10)** compulsory questions.
2. Show clearly all the working and answers in the spaces provided.
3. All working must be in blue or black ink except drawings which must be in pencil.
4. NECTA mathematical tables, Non programmable calculator geometric instruments and graph papers may be used where necessary.
5. Communication devices and any unauthorized materials are not allowed in the examination room.
6. Write your **Assessment Number** on the top right hand corner of every page of your answer booklet(s).

FOR ASSESSOR'S USE ONLY

QUESTION NUMBER	SCORE	ASSESSOR'S INITIALS
1		
2		
3		
7		
8		
9		
10		
TOTAL		
CHECKER'S INITIALS		

1. A student scored 18 out of 45 in a mathematics Test.

   a) Express this score as a fraction in its simplest form

   b) Convert the fraction in 2(a) above as a decimal.

c) What percentage did the student score?

d) If the pass mark is 50% did the student pass ?

2. a) Convert $1.\dot{2}1\dot{3}$ into fraction.

b) A famer has 1820 eggs collected from his poultry farm. He wants to pack them to trays of 30 eggs each.

i. How many full trays will he need?

ii. How many eggs will remain unpacked?

3. A student is making a rectangular vegetable garden of length 8m and width 5m.
   a) What's the area of the garden?

b) What's the perimeter of the garden?

c) If she wants to fence the garden using a wire that costs T.shs. 2,000/= per meter, how much money will she need?

4. a) You are given that the total cost of 3 pens and 2 pencils is 1,700 T.sh. One pen costs 400 T.shs.
   i. Let the cost of 1 pencil be x T.sh. Write an equation representing the total cost.

ii. Solve the equation to find the cost of one pencil.

b) Solve the following quadratic equation by using the general quadratic formula;

$10x^2 - 7x - 12 = 0$

5. a) John a form Two student at MtiMkavu secondary school had two plants seedling. He measured the rate at which the seedlings were growing. Seedling A grew 5cm in ten days, and seedling B grew 8cm in 12 days. What seedling was growing more quickly?

b) If x varies directly as the square of y, and x = 4 when y =2, find the value of x when y =8.

6. a) A line has a gradient of $\frac{3}{4}$ and passes through the point (-1,2). Find:

i. It's equation

ii. The y – intercept

b) An equation is defined by $y = 2x - 3$

i) Complete the table below for $x = -2, -1, 0, 1, 2$

x	-2	-1	0	1	2
y					

ii) Plot the graph by using the values of x and y from the table above.

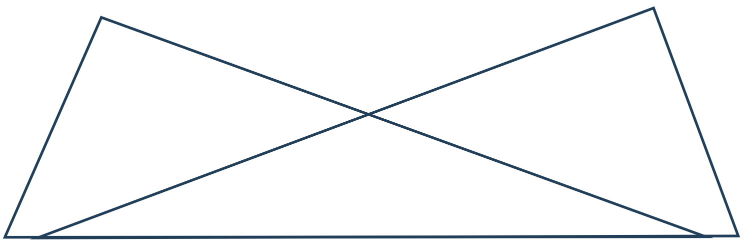
7. a) Given that P = $W\left(\frac{1+a}{1-a}\right)$, Make a the subject of the formula.

b) Rationalize the denominator of the expression

$$\frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$$

c) Find the value of x if $\log x = \log(3x - 5) - \log 2$

8. a) In the following figure, $\overline{AB} = \overline{DC}$ and $A\hat{B}C = D\hat{C}B$. Prove that $\Delta ABC \equiv \Delta DCB$



b) If triangle $\Delta ABC \sim \Delta\, PQR$ and $\overline{AC}$ = 20cm, $\overline{PR}$ = 10cm, $\overline{QR}$ = 12cm and $\overline{PQ}$ = 9cm. Find the lengths $\overline{AB}$ and $\overline{BC}$

9. (a) Without using mathematical tables find the exactly values of the following expressions:

   (i) $\sqrt{3}\tan 60°$

(ii) $\sin 60° \ (\tan 30° - \cos 30°)$

b) A rope of length 18 cm is tied to the top of a flagpole. The other end of the rope is fixed to a point 13 cm from the base of the flagpole. How high is the flagpole?

10. a) In a class of 40 students, 26 like football, 18 like basketball and 10 like both.

    i. Draw a Venn diagram to represent this information

    ii. How many students like football only?

iii. How many students like neither sport?

b) The examination marks of 45 students were as follows;

65	58	71	62	64	35	72	32	64
46	59	82	73	76	54	63	63	75
71	61	36	64	80	61	64	76	64
60	68	48	35	92	73	46	24	35
43	30	50	70	40	46	64	27	28

i). Construct a frequency distribution table using class interval 21 – 30, 31- 40, 41 – 50, and so on.

ii). Draw a cumulative frequency curve.

MATHEMATICS

VOCATIONAL STREAM

MARKING GUIDE

CSSC

2025

FORM II.

1. Given:

Marks scored by a student = 18.

Total marks of a test = 45

a. In fraction form.

$$= \frac{\text{Marks scored}}{\text{Total marks.}}$$ — 001 mark

$$= \frac{18}{45}$$ —— 001 mark

$$= \frac{2}{5}$$

∴ In fraction form is $2/5$. 001 mark

b. In decimal form.

$$\frac{2}{5} = 0.4$$ (002 marks)

$$\begin{array}{r} 0.4 \\ 5\overline{)\,2} \\ -0 \\ \hline 20 \\ -20 \\ \hline -- \end{array}$$

∴ In decimal form is 0.4 (001 mark)

c. Percentage of student's score $= \frac{\text{Marks scored}}{\text{Total mark}} \times 100\%$ 001 mark

$$= \frac{18}{45} \times 100\%$$

$$= \frac{2}{5} \times 100\%$$ 001 mark

$$= 40\%$$

∴ A student scored 40% in a test. -001 mark

d. Pass mark = 50%

Mark scored = 40%

Since 40% is less than 50%

∴ The student did not pass a test 001 mark

2\. a. Given: $1.\dot{2}1\dot{3}$

Let $x = 1.\dot{2}1\dot{3}$ ——— (i)

Multiply by 1000 both sides of the eqn (i) ——— 001 Mark

$1000x = 1213.213$ ——— (ii)

Substract eqn (i) from eqn (ii).

$$-\begin{cases} 1000x = 1213.\dot{2}1\dot{3} \\ x = 1.\dot{2}1\dot{3} \end{cases}$$ ——— 01 Mark

$$1000x - x = 1213.\dot{2}1\dot{3} - 1.\dot{2}1\dot{3}$$

$$\frac{999x}{999} = \frac{1212}{999}$$ ——— 001 Mark

$$x = \frac{1212}{999}$$

$$x = \frac{404}{333}$$ ——— 001 Mark

Since $x = 1.\dot{2}1\dot{3}$

$\therefore 1.\dot{2}1\dot{3} = \frac{404}{303}$. ——— 001 Mark

b. Given:

Total number of eggs $= 1820$.

Number of eggs a tray can carry $= 30$.

i. Number of full trays $= \frac{1820}{30}$ ——— 001 Mark

$= \frac{182}{3}$

$= 3 \overline{)182}$ quotient 60, -18, remainder $--2$ ——— 001 Mark

$\therefore$ Number of full trays needed is 60 trays. ——— 001 Mark

2. b. ii. Unpacked eggs total = Remainder from (b) (i) above — 00 1 mark

= 2 eggs.

$\therefore$ 2 eggs will remain unpacked. — 00 1 mark

3. <u>Given</u>:

Length $(l) = 8m$.

Width $(w) = 5m$.

a. Area of the garden $=$ Area of a rectangle — 001

$= \text{Length} \times \text{Width}$

$= 8m \times 5m$ — 001

$= 40m^2$.

$\therefore$ Area of the garden is $40m^2$. — 001

b. Perimeter of the garden $=$ Perimeter of a rectangle

$= 2(\text{Length} + \text{Width})$ — 001

$= 2(8m + 5m)$ — 001

$= 2(13m)$

$= 26m$.

$\therefore$ Perimeter of the garden is $26m$. — 001

c. Cost that will be used to fence the garden $= ?$

$=$ Perimeter of the garden $\times$ Cost of 1m. 001 Mark

$= 26m \times 2{,}000$ Tshs/m — 001 Mark

$= 52000$ Tshs. — 001 Mark

$\therefore$ Total cost that will be used to fence the garden is Tshs. 52,000/= — 001 Mark

4. Given:

a. Total cost of 3 pens and 2 pencils = 1,700 Tshs. — 001 Mark

Cost of one pen = 400 Tshs.

Cost of 3 pens $= 3 \times 400$ Tshs

Cost of 3 pens. = 1200 Tshs. — 001 Mark

x is the cost of 1 pencil.

Cost of 2 pencils $= 2x$.

i. In equation form: $1200 + 2x = 1700$. — — — — 001 Mark

ii. Solving the equation.

$$1200 + 2x = 1700$$

$$2x = 1700 - 1200$$

$$\frac{2x}{2} = \frac{500}{2}$$ — 001 Mark

$$x = \frac{500}{2}$$

$$x = 250$$

But x = Cost of 1 pencil.

∴ Cost of one pencil is 250 Tshs. — 001 Mark

b. Given: $10x^2 - 7x - 12 = 0$.

From the equation:

$a = 10$

$b = -7$

$c = 12$

Recall: Quadratic formula (General Quadratic formula).

$$x = \frac{+b \pm \sqrt{b^2 - 4ac}}{2a}$$ — 001 Mark

4\. b.

$$x = \frac{-(-7) \pm \sqrt{(-7)^2 - (4\times10\times{}^{-}12)}}{2\times10}$$

$$x = \frac{7 \pm \sqrt{49-({}^{-}480)}}{20}$$ — 01 Mark

$$x = \frac{7 \pm \sqrt{49+480}}{20}$$

$$x = \frac{7 \pm \sqrt{529}}{20}$$

$$x = \frac{7 \pm 23}{20}$$

$x = \dfrac{7+23}{20}$ or $x = \dfrac{7-23}{20}$ — 01 Mark

$x = \dfrac{30}{20}$ or $x = \dfrac{-16}{20}$

$x = \dfrac{3}{2}$ or $x = \dfrac{-4}{5}$ — 01 Mark

$x = 1.5$ or $x = -0.8$

$\therefore$ The value of x is 1.5 or -0.8. — 01 Mark

5. a. Rate of seedling A $= \dfrac{\text{Height of seedling A}}{\text{Time of growth.}}$ —— 01 | Mark

$= \dfrac{5\ cm}{10\ days}$ —— 01 | Mark

Rate of seedling A $=$ $\underline{0.5\ cm/days.}$

Rate of seedling B $= \dfrac{\text{Height of seedling B.}}{\text{Time of growth}}$ —— 01 | Mark

$= \dfrac{8\ cm}{12\ days.}$

Rate of seedling B $=$ $\underline{0.66\ldots = 0.67\ cm/day.}$ —— 01 | Mark

Compare Rate of seedling A and that of seedling B.

$0.5\ cm/day < 0.67\ cm/day$

$\therefore$ Seedling B was growing more quickly than seedling A. 01 | Mark

b. <u>Given:</u>

$X \propto y^2.$ —— 01 | Mark

$X = ky^2$

$k = \dfrac{x}{y^2}$ —— 01 | Mark

When $x = 4$; $y = 2.$

Then

$k = \dfrac{4}{(2)^2}$ —— 01 | Mark

$k = \dfrac{4}{4}$

$k = 1.$ —— 01 | Mark

So: $X = (1)\,y^2$

$X = y^2$

Required is x When $y = 8.$ —— 01 | Mark

5. b.

$$X = (8)^2$$

$$X = 64.$$

$\therefore$ $X = 64$ when $y = 8$. — — ½ mark

6. a. Given:

Slope of the line $(m) = 3/4$

Point $= (-1, 2)$

Let $(x_1, y_1) = (-1, 2)$

i. It's equation.

Recall the formula for finding equation of a line

$$y = m(x - x_1) + y_1.$$

$$y = \frac{3}{4}(x - (-1)) + 2.$$ —— 01 Mark

$$y = \frac{3}{4}(x+1) + 2.$$

$$4y = 3(x+1) + (2 \times 4)$$

$$4y = 3x + 3 + 8$$

—— 01 Mark

$$\frac{4y}{4} = \frac{3x}{4} + \frac{11}{4}$$

$$y = \frac{3}{4}x + \frac{11}{4}$$

∴ Equation of a line is $y = \frac{3}{4}x + \frac{11}{4}$ —— 01 Mark

ii. From the equation obtained in (a) (i) above:

$$y = \underbrace{\frac{3}{4}}_{\text{slope}}x + \underbrace{\frac{11}{4}}_{y\text{-intercept}}$$ —— 01 Mark

∴ y-intercept is $\frac{11}{4}$ —— 01 Mark

6\. a. (ii)

ALTERNATIVELY.

y - intercept is obtained when the value of $x = 0$.

So:

$$y = \frac{3}{4}x + \frac{11}{4}$$

$$y = \frac{3}{4}(0) + \frac{11}{4}$$

$$y = 0 + \frac{11}{4}$$

$$y = \frac{11}{4}$$

$\therefore$ y - intercept is $\frac{11}{4}$

b. Given: $y = 2x - 3$.

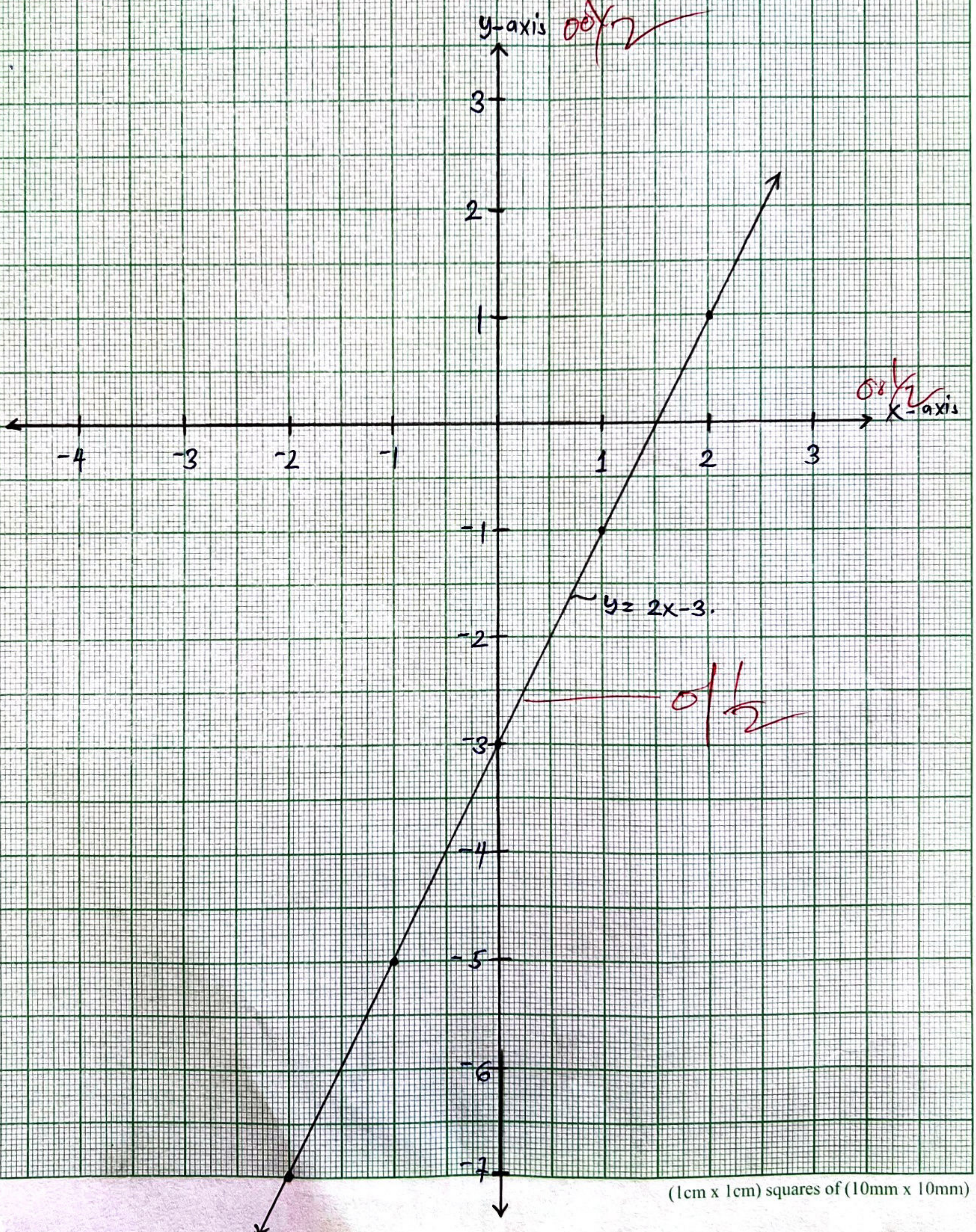
i. Table of values.

x	-2	-1	0	1	2
y	-7	-5	-3	-1	1

6.(b)
(ii)

A GRAPH OF EQUATION $y = 2x - 3$.

SCALE
V.S 2 cm represent 1 unit
H.S 2 cm represent 1 unit



(1cm x 1cm) squares of (10mm x 10mm)

7. a. <u>Given:</u>

$$P = W\left(\frac{1+a}{1-a}\right)$$

Divide by W both sides.

$$\frac{P}{W} = \frac{W}{W}\left(\frac{1+a}{1-a}\right)$$

$$\frac{P}{W} \times \left(\frac{1+a}{1-a}\right)$$ — 01 mark

$$W(1+a) = P(1-a)$$

$$W + Wa = P - Pa.$$

$$Wa + Pa = P - W$$

$$\frac{a(W+P)}{W+P} = \frac{P-W}{W+P}$$

$$a = \frac{P-W}{W+P}$$ — 01 mark

or

$$a = \frac{P-W}{P+W}.$$

$\therefore$ $a = \dfrac{P-W}{W+P}$ or $a = \dfrac{P-W}{P+W}.$ — 01 mark

b. Given: $\dfrac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$

$$= \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}} \times \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}+\sqrt{2}}$$ — 01 mark

7. b.

$$= \frac{\sqrt{9} + \sqrt{6} + \sqrt{6} + \sqrt{4}}{\sqrt{9} + \sqrt{6} - \sqrt{6} - \sqrt{4}}$$

$$= \frac{\sqrt{9} + 2\sqrt{6} + \sqrt{4}}{\sqrt{9} - \sqrt{4}}$$ —— 01 Mark

$$= \frac{3 + 2\sqrt{6} + 2}{3 - 2}$$

$$= \frac{5 + 2\sqrt{6}}{1}$$

$$\therefore \frac{\sqrt{3} + \sqrt{2}}{\sqrt{3} - \sqrt{2}} = 5 + 2\sqrt{6}.$$ —— 01 Mark

c. If: $\log x = \log (3x - 5) - \log 2.$

$$\log x = \log \left(\frac{3x - 5}{2}\right)$$

$$x = \frac{3x - 5}{2}$$ —— 01 Mark

$$2x = 3x - 5$$ —— 01 Mark

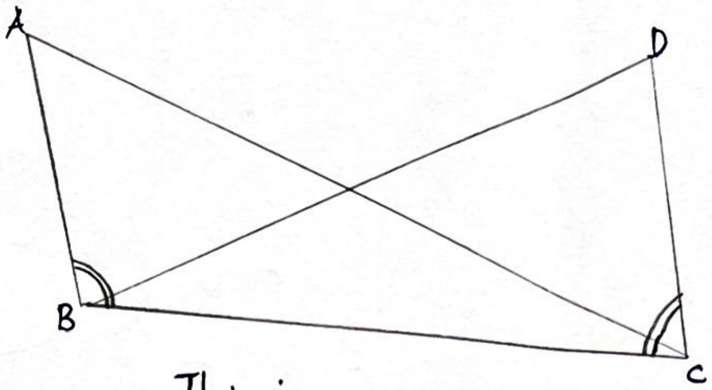
$$2x - 3x = -5$$

$$\frac{-x}{-1} = \frac{-5}{-1}$$ —— 01 Mark

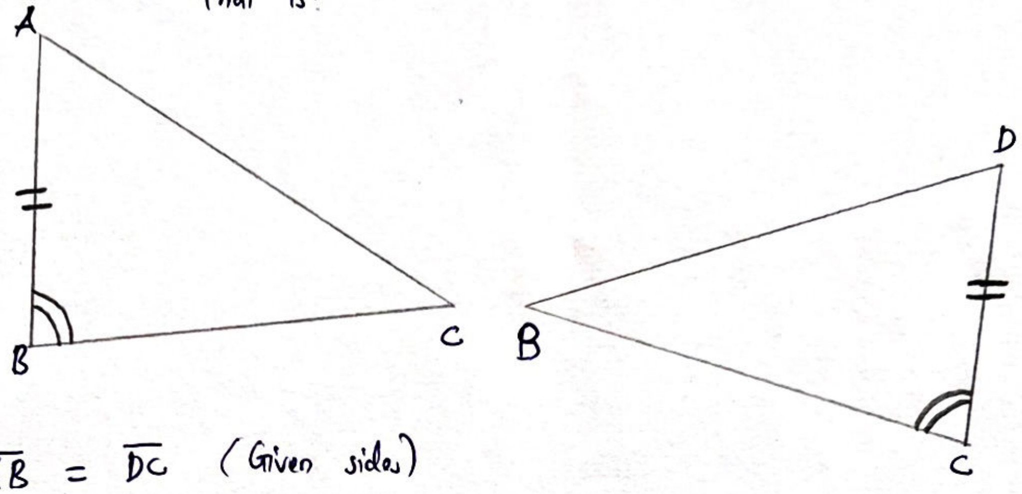
$$x = 5$$

$\therefore$ The value of x is 5. —— 01 Mark

8. a. <u>Given:</u>



That is:



i. $\overline{AB} = \overline{DC}$ (Given sides)

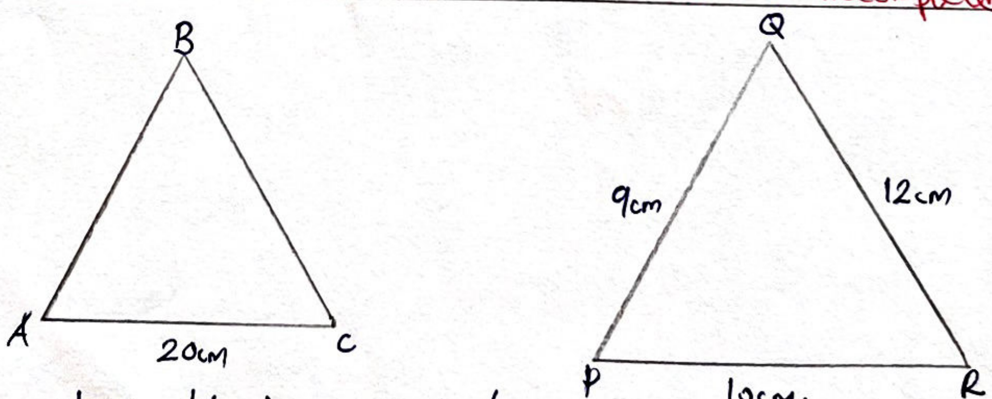
ii. $\overline{BC} = \overline{BC}$ (Common sides)

iii. $A\hat{B}C = D\hat{C}B$ (Given Angle).

$\therefore$ $\triangle ABC \equiv \triangle DCB$ by SAS.

skipped
due to incomplete ans.

b.



If the above triangles are similar.

$$\frac{\overline{AC}}{\overline{PR}} = \frac{\overline{AB}}{\overline{PQ}} = \frac{\overline{BC}}{\overline{QR}}$$ 00

For $\frac{20cm}{10cm} = \frac{\overline{AB}}{9cm}$ 00

Then.

8. b.

$$\frac{9\text{cm} \times 20\text{cm}}{10\text{cm}} = \overline{AB}$$

$$\overline{AB} = 18\text{cm}$$

Also:

$$\frac{\overline{AB}}{\overline{PQ}} = \frac{\overline{BC}}{\overline{QR}}$$

$$\frac{18\text{cm}}{9\text{cm}} = \frac{\overline{BC}}{12\text{cm}}$$

$$\overline{BC} = \frac{18\text{cm} \times 12\text{cm}}{9\text{cm}}$$

$$\overline{BC} = 24\text{cm}.$$

$\therefore$ The lengths of $\overline{AB}$ and $\overline{BC}$ are 18cm and 24 cm respectively.

9. a.

i. Given: $\sqrt{3}\tan 60°$.

But: $\tan 60° = \sqrt{3}$

Then

$$\sqrt{3}\tan 60° = \sqrt{3}\cdot\sqrt{3}$$ ——— 80

$$= \sqrt{9}$$ ——— 00

$$= 3.$$

$\therefore \sqrt{3}\tan 60° = 3.$ ——— 00 1/2

ii. $\sin 60°(\tan 30° - \cos 30°)$.

But:

$\sin 60° = \frac{\sqrt{3}}{2}$.

$\tan 30° = \frac{\sqrt{3}}{3}$ ——— 00 1 1/2 Marks

$\cos 30° = \frac{\sqrt{3}}{2}$.

Then:

$$\sin 60°(\tan 30° - \cos 30°) = \frac{\sqrt{3}}{2}\left(\frac{\sqrt{3}}{3} - \frac{\sqrt{3}}{2}\right)$$ ——— 00 1 Mark

$$= \frac{\sqrt{3}}{2}\left(\frac{2\sqrt{3} - 3\sqrt{3}}{6}\right)$$

$$= \frac{\sqrt{3}}{2}\left(\frac{-\sqrt{3}}{6}\right)$$ ——— 00 1/2 Mark

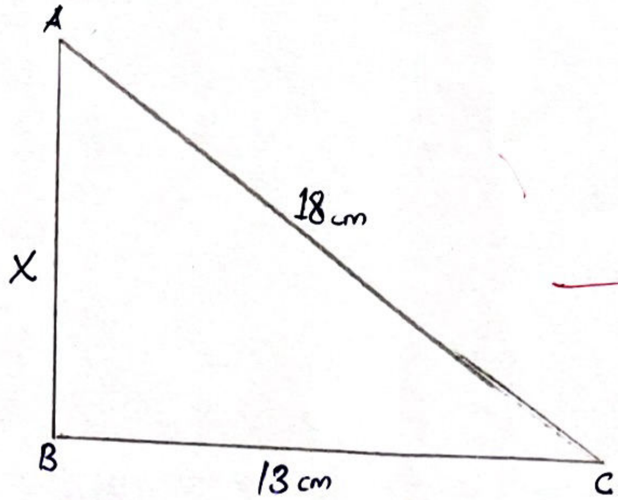
$$= \frac{-\sqrt{9}}{12}$$

$$= \frac{-3}{12}$$

$$= \frac{-1}{4}$$

$\therefore \sin 60°(\tan 30° - \cos 30°) = \frac{-1}{4}$ ——— 00 1/2 Mark

9. b. From the question.



00 | Mark

By using Pythagoras theorem.

$(BC)^2 + (AB)^2 = (AC)^2$ —— 00 | Mark

$(13\,cm)^2 + (X)^2 = (18\,cm)^2$ —— 00 | Mark

$169\,cm^2 + X^2 = 324\,cm^2$

$X^2 = 324\,cm^2 - 169\,cm^2$ —— 00 | Mark

$X^2 = 155\,cm^2$

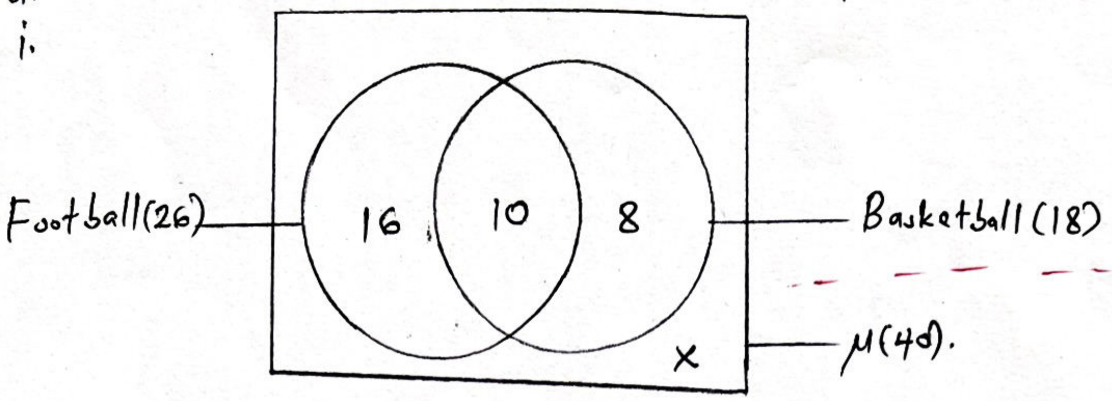
$X = \sqrt{155\,cm^2}$ —— 00 | Mark

$X = 12.45\,cm.$

$\therefore$ The length of the flagpole is 12.45 cm. —— 00 | Mark

10. a. Let: Football be F and Basketball be B.

i.



002

ii. Student who like football only = n(Football) − n(F∩B)

$= 26 - 10$

$= 16$ students

∴ 16 students like football only. 001

iii. Student who like neither of the sports.

Recall:

$\mu = n(F)_{only} + n(B)_{only} + n(F \cap B) + n(F \cap B)'$ 001

$40 = 16 + 8 + 10 + x$

$40 = 34 + x$ 001

$x = 40 - 34$

$x = 6$ students.

∴ 6 students like neither of the sports. 001

10. b.

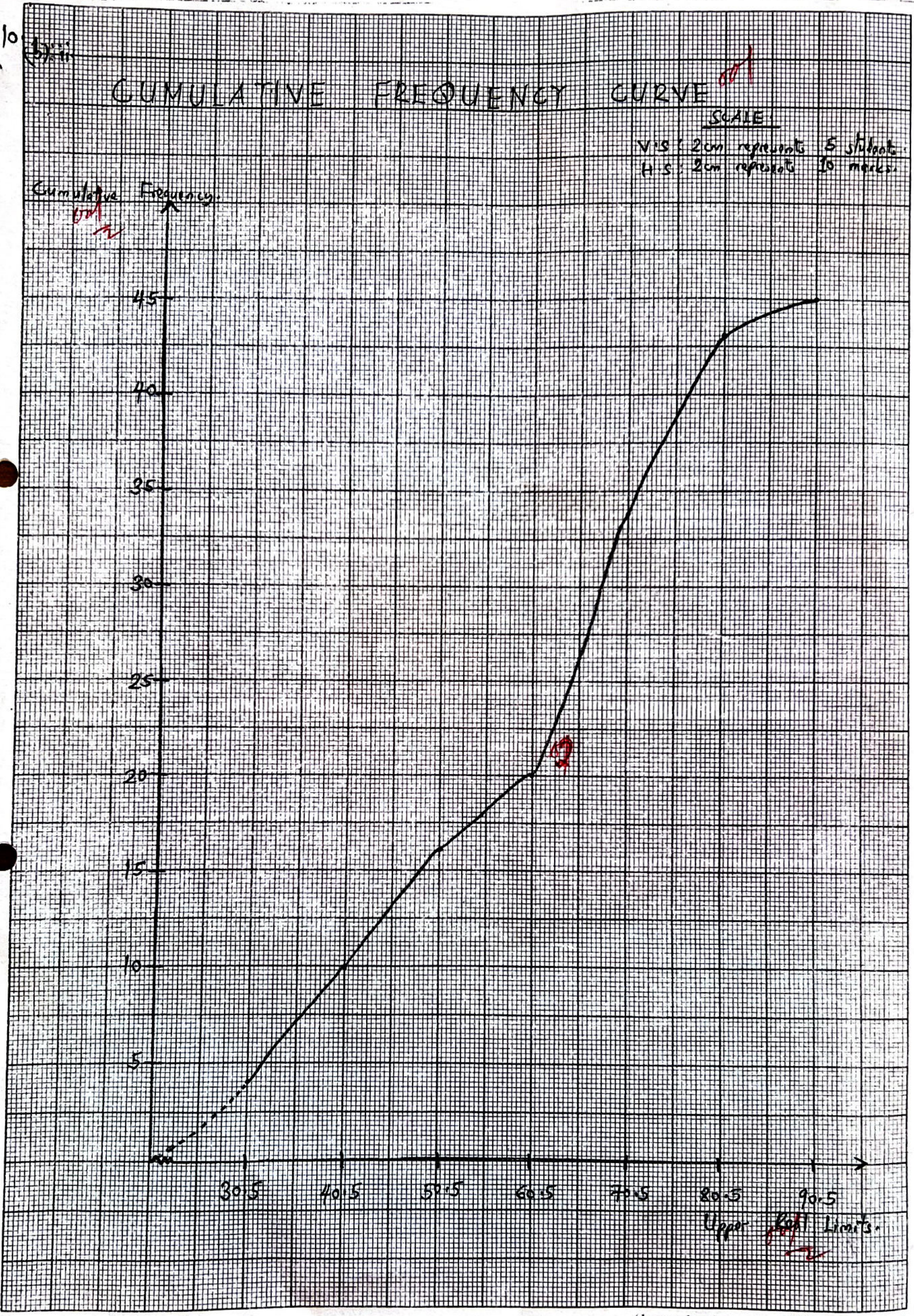
i.

FREQUENCY DISTRIBUTION TABLE.

Class Interval	Frequency	C.F	Upper Real Limits.
21 - 30	4	4	30.5
31 - 40	6	10	40.5
41 - 50	6	16	50.5
51 - 60	4	20	60.5
61 - 70	14	34	70.5
71 - 80	9	43	80.5
81 - 90	2	45	90.5

a½ a½ a½ a½

10 (b)(ii)



(1cm x 1cm) squares of (10mm x 10mm)